

KETO FILM COATED TABLET 10 MG

Ingredient(s):

Each tablet contains:

Ketorolac Tromethamine 10 mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Pharmacodynamic

1. Ketorolac is a nonsteroidal agent with potent analgesic, anti-inflammatory and antipyretic activity. It inhibits the synthesis of prostaglandins in body tissues by inhibiting cyclooxygenase, an enzyme that catalyzes the formation of prostaglandin precursors (endoperoxides) from arachidonic acid. Ketorolac does not have any known effects on opiate receptors.

Pharmacokinetic

1. Ketorolac is completely absorbed following oral administration with a mean peak plasma concentration of 0.7~1.1 µg/mL occurring an average of 44 minutes after a single 10mg dose in fasted subjects. The terminal plasma half-life is 2.4-9.0 hours in young adults and 4.3~7.6 hours in elderly subjects (mean age 72).
2. The primary route of excretion of ketorolac and its metabolites (conjugates and the p-hydroxy metabolite) is in the urine (91.4%) with the remainder (6.1%) being excreted in the feces. Ketorolac is almost totally bound to plasma proteins (more than 99%) and its apparent volume of distribution is 0.25 L/kg or less.
3. Several clinical studies indicated that ketorolac 10mg to 30mg orally compares favourably with other oral analgesic agents including aspirin, paracetamol, naproxen sodium, glafenine and ibuprofen, and with combinations of opioid and non-steroidal or simple analgesics. In comparative trials, ketorolac caused fewer adverse events and was discontinued less often than either morphine or meperidine. Besides, ketorolac did not affect patient's psychomotor activity and respiratory function, and caused less constipation and gastric mucosal injury than oral Aspirin.

Indication(s):

Short term management of moderate to severe acute post-operative pain following surgical procedures associated with low risk of haemorrhage.

Dosage and Administration:

Adult over 16 years old : 10mg every 4-6 hours.

Elderly ≥ 65 years old : 10mg every 6-8 hours.

Maximum 40 mg daily. Duration of treatment should not exceed 7 days.

When converting from parenteral to oral administration, total combined dose on the day of converting should not exceed 90mg (60mg in elderly and patients weighing less than 50kg) of which the oral component should not exceed 40mg. To be dispensed on physician's prescription.

After assessing the risk/ benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

Contraindications:

1. History of hypersensitivity to ketorolac, aspirin as severe anaphylactic reaction reported.
2. History of asthma, complete or partial syndrome of nasal polyps, angioedema or bronchospasm.
3. History of peptic ulceration or gastrointestinal bleeding.
4. History of haemorrhagic diatheses (including coagulation disorder) and operations with high risk of haemorrhage or incomplete haemostasis.
5. Confirmed or suspected cerebrovascular bleeding.
6. Moderate or severe renal impairment.
7. Hypovolaemia or dehydration.
8. Pregnancy, delivery or lactation.
9. Children < 16 years of age.
10. Patients currently receiving other NSAIDs, oxpentifylline, probenecid, or lithium salts.
11. Patients on anticoagulant including low dose heparin (2500-5000 units twelve hourly).

Precautions(s) / Warning(s):

1. Ketorolac is a potent inhibitor of prostaglandin synthesis. Reduction in renal blood flow has been observed as prostaglandins have a supportive role in the maintenance of renal perfusion. Patients with underlying renal insufficiency are at increased risk of developing acute renal failure, the risk and benefits should be assessed prior to giving ketorolac to these patients.
2. Ketorolac may cause elevations of liver enzymes. In patients with pre-existing liver dysfunction, it may lead to the development of a more severe hepatic reaction.
3. All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulceration occurs in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patients about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.
4. Anaphylactoid reactions may occur in individuals with a history of angioedema, bronchospasm reactivity (eg. asthma) and nasal polyps. Anaphylactoid reactions, like anaphylaxis, may have a fatal outcome.
5. Fluid retention and oedema have been observed in some patients taking NSAIDs, therefore caution is advised in patients with fluid retention or heart failure.
6. Ketorolac inhibits platelet aggregation and may prolong bleeding time; therefore, it is contraindicated as a preoperative medication, and caution should be used when hemostasis is critical.
7. Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose or duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.

There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAID use.
8. NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking anti-hypertensive with NSAIDs may have an impaired anti-hypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.
9. NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens-Johnson Syndrome (SJS), which can be fatal and

without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reaction and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.

WARNINGS

RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (eg. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (eg. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Drug Interaction :

1. Concurrent high-dose therapy with methotrexate may cause fatal interaction, probably because of delayed excretion of methotrexate and its resultant toxic levels.
2. Ketorolac may cause a degree of fluid retention and may reduce the efficacy of diuretics.
3. Ketorolac reduces clearance of lithium, resulting in higher blood levels.
4. Ketorolac may alter the binding of anticoagulants to human plasma protein.
5. Ketorolac may antagonize hypotensive effect of ACE inhibitors and can cause increased risk of renal damage and hyperkalaemia.
6. Concomitant administration of ketorolac with probenecid; oxpentifylline; and other NSAIDs will result in significant increased ketorolac plasma levels.

Side Effect(s) / Adverse Reactions(s):

Side-effects reported include anaphylaxis (with rash, bronchospasm, laryngeal oedema and hypotension), fluid retention, nausea, abdominal discomfort, bowel changes, peptic ulceration, gastro-intestinal bleeding (elderly at greater risk), pancreatitis, drowsiness, dizziness, headache, sweating, dry mouth, excessive thirst, mental and sensory changes, convulsions, myalgia, aseptic meningitis, hyponatraemia, hyperkalaemia, raised blood urea and creatinine, urinary symptoms and acute renal failure, flushing or pallor, bradycardia, hypertension, purpura, thrombocytopenia, prolonged bleeding time, dyspnoea and pulmonary oedema, skin reactions (some severe, including Stevens-Johnson and Lyell's syndromes), postoperative wound haemorrhage, haematoma, epistaxis, oedema, and liver function changes.

Symptoms and Treatment For Overdosage, and Antidote(s):

Symptoms of overdosage include abdominal pain and peptic ulcers which healed after discontinuation of dosing. Metabolic acidosis has been reported following intentional overdosage. Although there is lack of experience with ketorolac overdosage, recommended treatment consist of the following:

- Carefully monitoring the patient and instituting treatment of observed symptoms.
- For renal failure – Dialysis may be needed. However, dialysis is not likely to assist in removing ketorolac from the body after an overdose; decrease clearance and prolongation of half-life have been reported in patients receiving dialysis.

Route of Administration : To be taken orally.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Shelf-life:

3 years from the date of manufacture.

Product description and Packing(s):

A white color film coated tablet; one side impressed with the mark "N", and the other side impressed with "YSP".



Blister packing of 10's x 10 and 10's x 50.

Bahan Aktif:

Setiap tablet mengandungi:

Ketorolac Tromethamine 10 mg

Indikasi:

Untuk melegakan kesakitan yang sederhana ke teruk selepas pembedahan di mana risiko pendarahan adalah rendah.

Dos:

Dewasa berumur 16 tahun ke atas : 10mg setiap 4-6 jam.

Pesakit berumur melebihi 65 tahun : 10mg setiap 6-8 jam.

Maksima 40mg sehari. Tempoh rawatan tidak harus melebihi 7 hari



Manufacturer and Product Registration Holder:
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